

JOSUE TORRES-FONSECA

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EDUCATION

Doctor of Philosophy in Computer Science and Engineering

University of Michigan, Ann Arbor, MI

Expected May 2028

GPA: 4.0

M.S. Computer Science and Engineering,

University of Michigan, Ann Arbor, MI

Graduated May 2025

GPA: 4.0

B.S. Computer Science, Minor: Applied Mathematics

Boise State University, Boise, ID

Graduated May 2023

GPA 3.889; Magna Cum Laude

QUALIFICATIONS

- Computer programming in proficiency in Python and Java
- Working knowledge of C#, SQL, Javascript and Git
- Effective problem solving and debugging skills
- Knowledgeable on data structures and algorithms
- Experience with machine learning libraries in python including PyTorch, Numpy, and Pandas
- Experience with Vision Language Models and Large Language Models
- Experience with agile development
- Ability to work on a team and effective communication skills

WORK EXPERIENCE

University of Michigan

August 2023 – Present

Supervised by Dr. Joyce Chai, Graduate Researcher

- Created a safety-centric dataset using PDDL to generate trajectories that demonstrate how to complete household tasks while mitigating encountered hazards rendered within the AI2-Thor simulator
- Evaluated 11 state-of-the-art VLMs on their ability to recognize hazards via visual question answering and generate safe plans to complete household tasks through proactive hazard mitigation within the A12-THOR simulator
- Developed a decoupled multi-agent framework that separated hazard recognition from task execution, resulting in an increase in safety-planning success rates
- Accepted to Findings of the Association for Computational Linguistics 2026

Boise State University

August 2021 – August 2023

Supervised by Dr. Casey Kennington, Undergraduate Research Assistant

- Published and presented two papers at Special Interest Group on Discourse and Dialogue 2022 and Language Resources and Evaluation Conference 2022
- Designed and built an incremental spoken dialogue system capable of transcribing and understanding speech, detecting, and learning about objects, and mapping objects in its environment using SLAM
- Maintained and controlled robots using a Python SDK
- Gathered, organized and analyzed data from research participants

Massachusetts Institute of Technology

May 2022 – November 2022

Supervised by Dr. Jacob Andreas, Undergraduate Research Assistant

- Helped mentor with natural language processing research regarding interpretable machine learning
- Created new MILAN pipeline that requires less training data
- Prepared presentations about findings and present to other interns and faculty

Paylocity

May 2021 – August 2021

Software Engineer Intern

- Contributed to the development of software through the software lifecycle as a part of a 13-member team
- Worked with product owners and leads to understand use cases and develop them into features

- Designed, developed, tested and deployed new features in project releases
- Leveraged and obtained familiarity with software development methodologies: Agile and Scrum
- Leveraged and obtained knowledge in Full Stack Web development including Javascript, React/Redux, C#, .Net framework, SQL and Git

Boise State University

August 2020 – May 2021

Tutor

- Tutored for Data Structures (Fall 2020), Introduction to Theory of Computation (Spring 2021), and Operating Systems (Spring 2023)
- Worked with students to enhance their understanding of topics discussed in class
- Debugged code quickly and efficiently
- Ensured all students understand their assignments and the "why" behind their code
- Built professional and peer relationships with staff and students respectively

Texas State University

June 2020 – August 2020

Supervised by Dr. Mina Guirguis, Undergraduate Research Assistant

- Helped mentor with research about using game theory to protect cyber physical systems
- Programmed in Python to help set up the environment needed to gather data
- Prepared research paper and presentations to introduce final findings to REU participants and organizers

Boise State University

August 2020 – December 2020

Lab Assistant

- Helped students with electrical engineering labs and answered questions
- Assisted professor in grading papers

ORGANIZATIONS

Treasurer, Computer Science and Engineering Graduate Students

August 2024 – May 2025

- Managed organization funds
- Assisted in organizing events

President, AI Club

August 2021 – May 2023

- Facilitated a variety of professional development and technical development workshops such as; python, machine learning, and data science
- Gathered and managed club funds

President, Competitive Cybersecurity and Cyber-Physical Club

August 2019 – May 2022

- Participated in CyberForce Competition, at INL (46th out of 103 teams) (Fall 2019)
- Presented on cybersecurity topics such as data scraping, and reverse engineering with Ghidra
- Lead Capture the Flag activities to help members gain foundational cybersecurity knowledge

HONORS & AWARDS

Gemini Academic Program Award	(\$10,000)	Spring 2026 – Spring 2027
NSF Graduate Research Fellowship	(\$159,000)	Fall 2023 – Spring 2028
Dean's List		Fall 2018 – Spring 2023
Boise State University Top Ten Scholar		Spring 2023
Barry M. Goldwater Scholarship	(\$7,500)	Fall 2022 – Spring 2023
Federal Employee Education & Assistance Fund Scholarship	(\$1,000)	Fall 2018